

```
import numpy as np
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score
```

```
df=pd.read_csv("pro.csv")
print(df)
print(type(df))
```

```
x=df[["problems_solved"]]
print(x)
y=df[["result"]]
print(y)
```

```
   problems_solved  result
0                40       0
1                50       0
2                70       1
3                80       1
4               100       1
<class 'pandas.core.frame.DataFrame'>
   problems_solved
0                40
1                50
2                70
3                80
4               100
   result
0        0
1        0
2        1
3        1
4        1
```

```
x_train,x_test,y_train,y_test=train_test_split(x,y,random_state=42)
print(x_train)
print(type(x_train))
print(y_train)
print(type(y_train))
```

```
   problems_solved
2                70
0                40
3                80
<class 'pandas.core.frame.DataFrame'>
   result
2        1
0        0
3        1
<class 'pandas.core.frame.DataFrame'>
```

```
lor=LogisticRegression()
lor.fit(x_train,y_train)
```

C:\ProgramData\anaconda3\Lib\site-packages\sklearn\utils\validation.py:1339: DataConversionWarning: A column-vector y was passed wh
y = column_or_1d(y, warn=True)

```
▼ LogisticRegression ⓘ ?
LogisticRegression()
```

```
y_pred=lor.predict(x_test)
print(y_pred)
```

```
[0 1]
```

```
accuracy=accuracy_score(y_test,y_pred)
print(accuracy*100)
```

```
100.0
```

```
probs = lor.predict_proba(x_test)
print(probabilities)
```



```
threshold = 0.5
predicted_class = (probs[:,1] >= threshold)
i = 1
if predicted_class[i] == 1:
    print(f"Predicted probability for class 1: {probs[i, 1]}")
```

```
[[8.19562128e-01 1.80437872e-01]
 [9.65859089e-07 9.99999034e-01]]
Predicted probability for class 1: 0.9999990341409107
```

```
f=(1/(1+
```